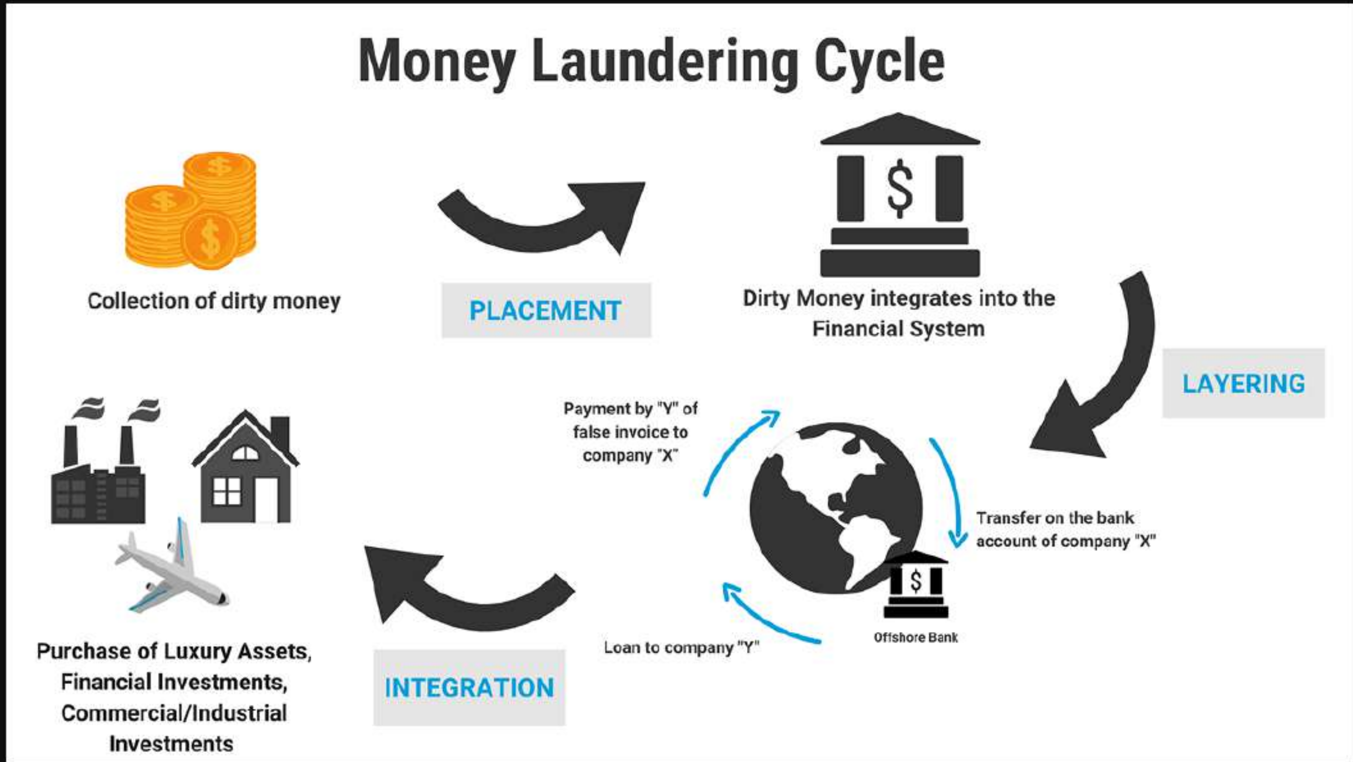
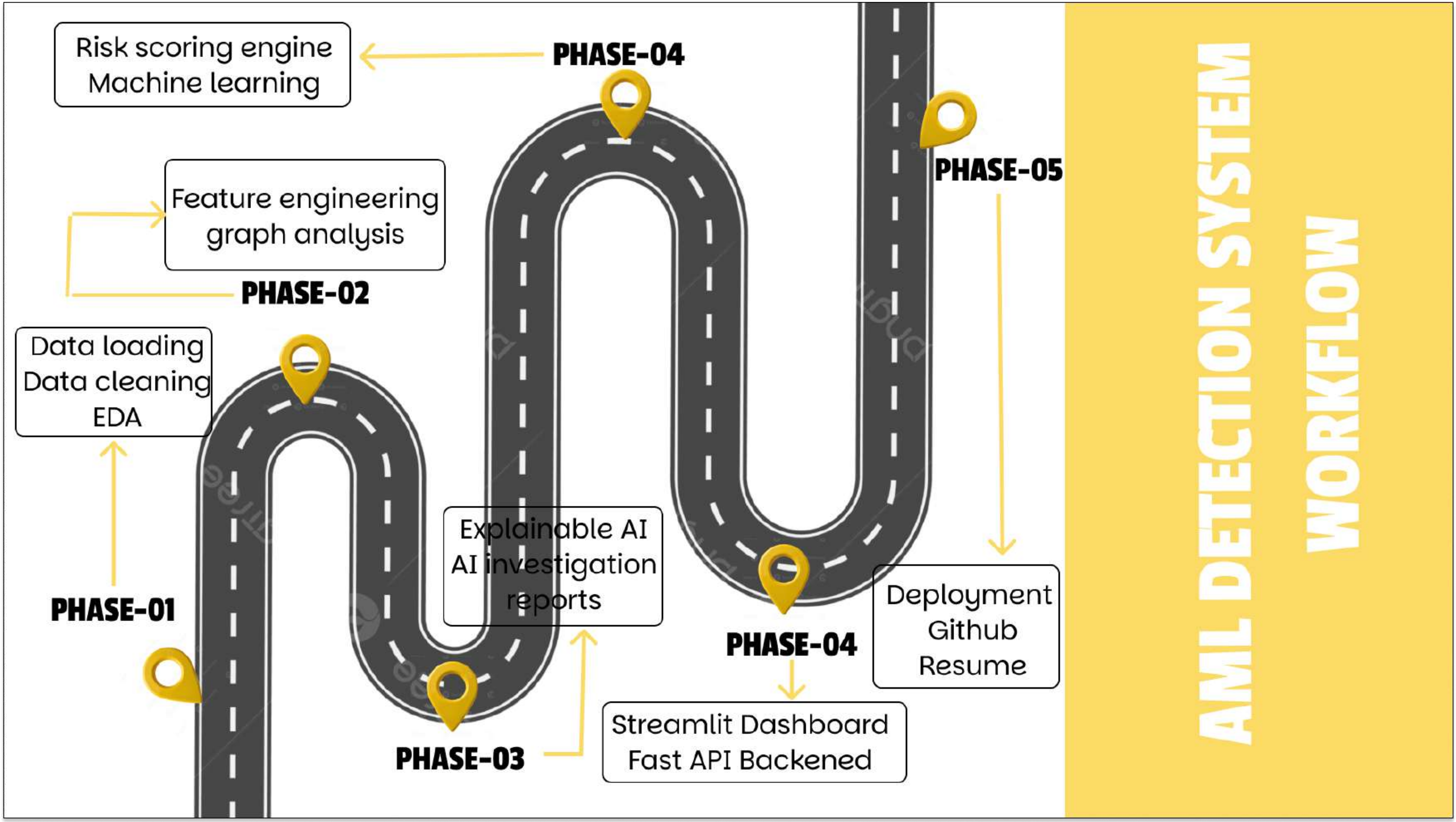
FINSHIELD

AI-powered aml detectection sytem

PROBLEM STATEMENT

Financial institutions handle a massive volume of daily transactions, making it difficult to manually detect suspicious activities. Existing rule-based Anti-Money Laundering (AML) systems generate a high number of false positives and often fail to identify complex and evolving money laundering patterns. There is a need for an intelligent, automated system that can accurately detect suspicious transactions and assist compliance teams in real time.



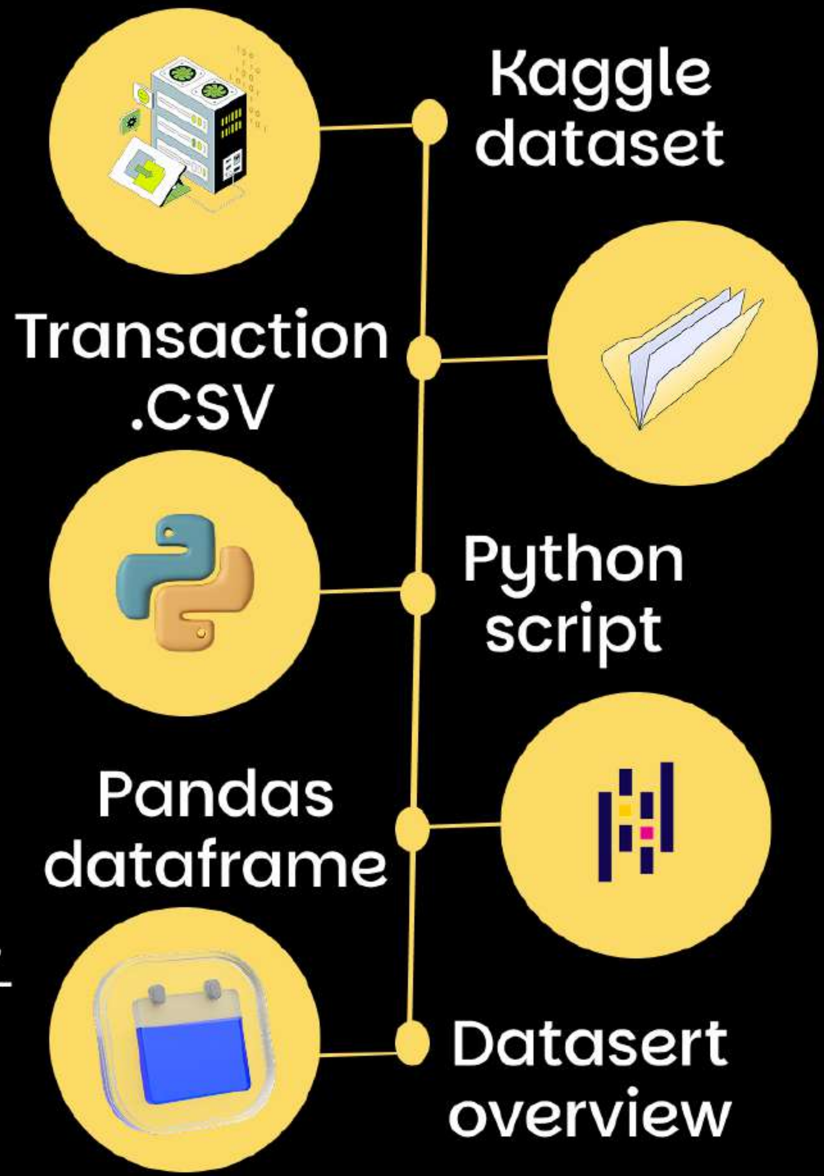

AML DETECTION SYSTEM WORKFLOW
PHASE-01
Data loading
Data cleaning
EDA
PHASE-02
Feature engineering
graph analysis
PHASE-03
Explainable AI
AI investigation reports
PHASE-04
Risk scoring engine
Machine learning
PHASE-04
Streamlit Dashboard
Fast API Backened
PHASE-05
Deployment
Github
Resume

DATASET COLLECTION AND LOADING

DATASET USED

Kaggle IBM AML Transaction dataset

"The dataset contains customer and transaction records, including transaction amounts, account details, and risk-related features used to detect money laundering activities."



DATA CLEANING AND PREPROCESSING

DATA INSPECTION

- Analyzed dataset structure
- Reviewed features and data types
- Examined statistical summaries

MISSING AND DUPLICATE VALUE CHECK

- Checked for missing values
- Identified duplicate records
- Removed inconsistencies where necessary

DATA TYPE CONVERSION

- Converted timestamp fields into datetime format
- Standardized feature format

DISTRIBUTION ANALYSIS

- Studied transaction amounts
- Analyzed payment methods
- Examined currency distribution
- Investigated laundering label distribution

OUTLIER DETECTION

- Detected unusually large transactions
- Flagged suspicious values instead of removing them

BEFORE

- Raw transaction records
- Inconsistent formats
- Potential duplicates
- Unverified data quality

AFTER

- Structured dataset
- Standardized features
- Validated records
- ML-ready data

EXPLORATORY DATA ANALYSIS

Under covering hidden transaction patterns

OBJECTIVE

"Analyze transaction behavior and identify patterns associated with money laundering activities before model development."

🔍 Are laundering transactions larger than normal transactions?

🔍 Which payment methods show higher risk?

🔍 Do suspicious transactions occur at specific times?

🔍 How imbalanced is the dataset?

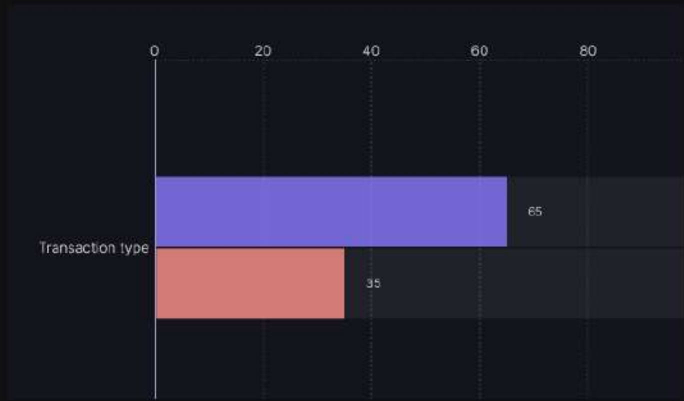
🔍 Which accounts exhibit unusual activity patterns?

KEY TRANSACTION INSIGHTS

CLASS DISTRIBUTION

1.It shows Normal Transactions vs Laundering Transactions
2.AML is an imbalanced classification problem.

TRANSACTION TYPE COMPARISON



Laundering transactions tend to involve larger amounts.

PAYMENT TYPE RISK

Wire transfers exhibit higher laundering rates.

SUSPICIOUS HOURS

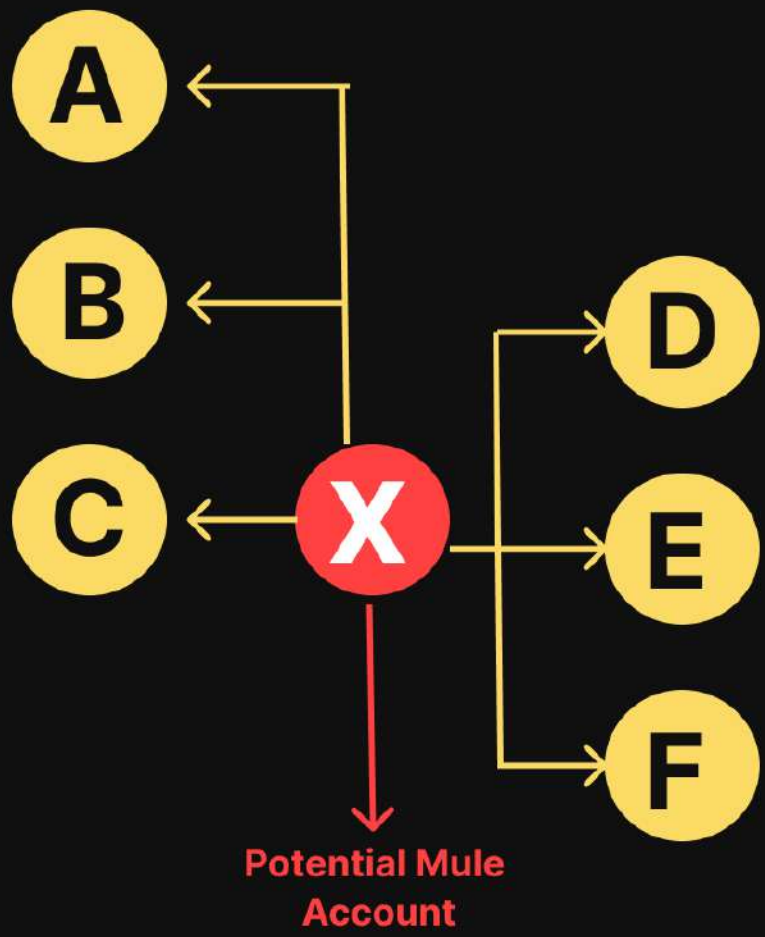
Suspicious activity peaks during late-night hours.

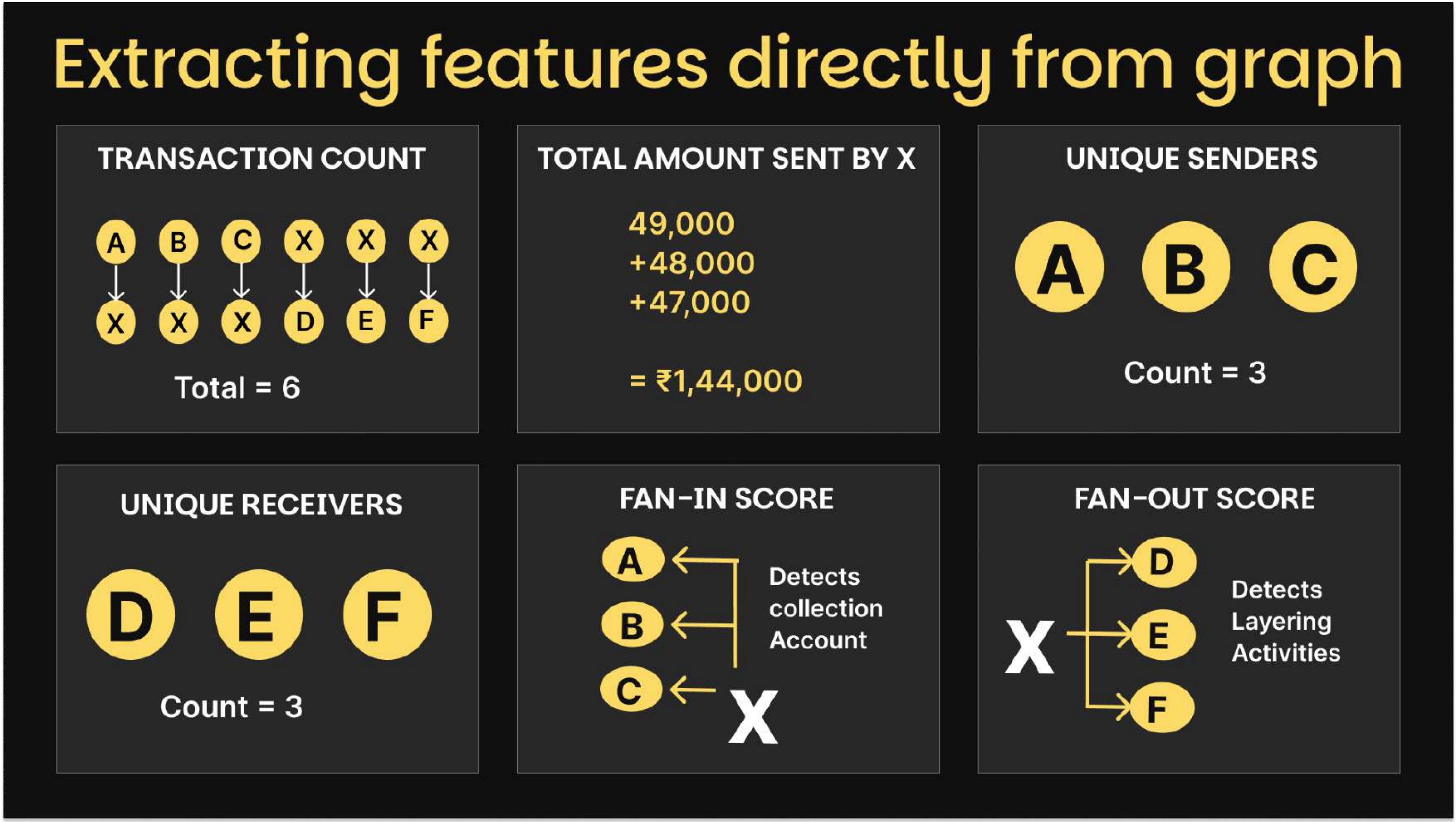
FEATURE ENGINEERING

EXAMPLE DATASET

SENDER	AMOUNT	RECEIVER
A	Rs. 50,000	X
B	Rs. 45,000	X
C	Rs. 55,000	X
X	Rs. 49,000	D
X	Rs. 47,000	E
X	Rs. 47,000	F

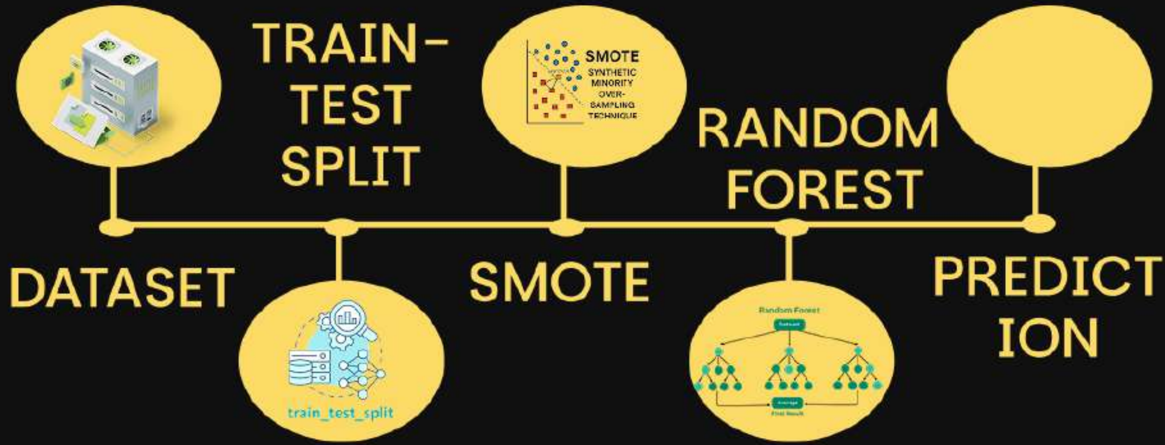
NETWORK GRAPH




Extracting features directly from graph
TRANSACTION COUNT
A B C X X X
X X X D E F
Total = 6
TOTAL AMOUNT SENT BY X
49,000
+48,000
+47,000
= ₹1,44,000
UNIQUE SENDERS
A B C
Count = 3
UNIQUE RECEIVERS
D E F
Count = 3
FAN-IN SCORE
A
B
C
X
Detects collection Account
FAN-OUT SCORE
X
D
E
F
Detects Layering Activities

MACHINE LEARNING MODEL

ML PIPELINE



INPUT FEATURES

FEATURE CATEGORY	EXAMPLE
Transaction features	transaction count,avg count
Network features	fan-in,fan-out,centrality
Risk features	Risk score
Time features	Nignt transaction

MODEL EVALUATION

PERFORMANCE METRICS

PRECISION
How many flagged accounts are actually suspicious?

RECALL
How many laundering accounts were detected?

F1 SCORE
Balance between Precision and Recall

CONFUSION MATRIX
Compare actual vs predicted results

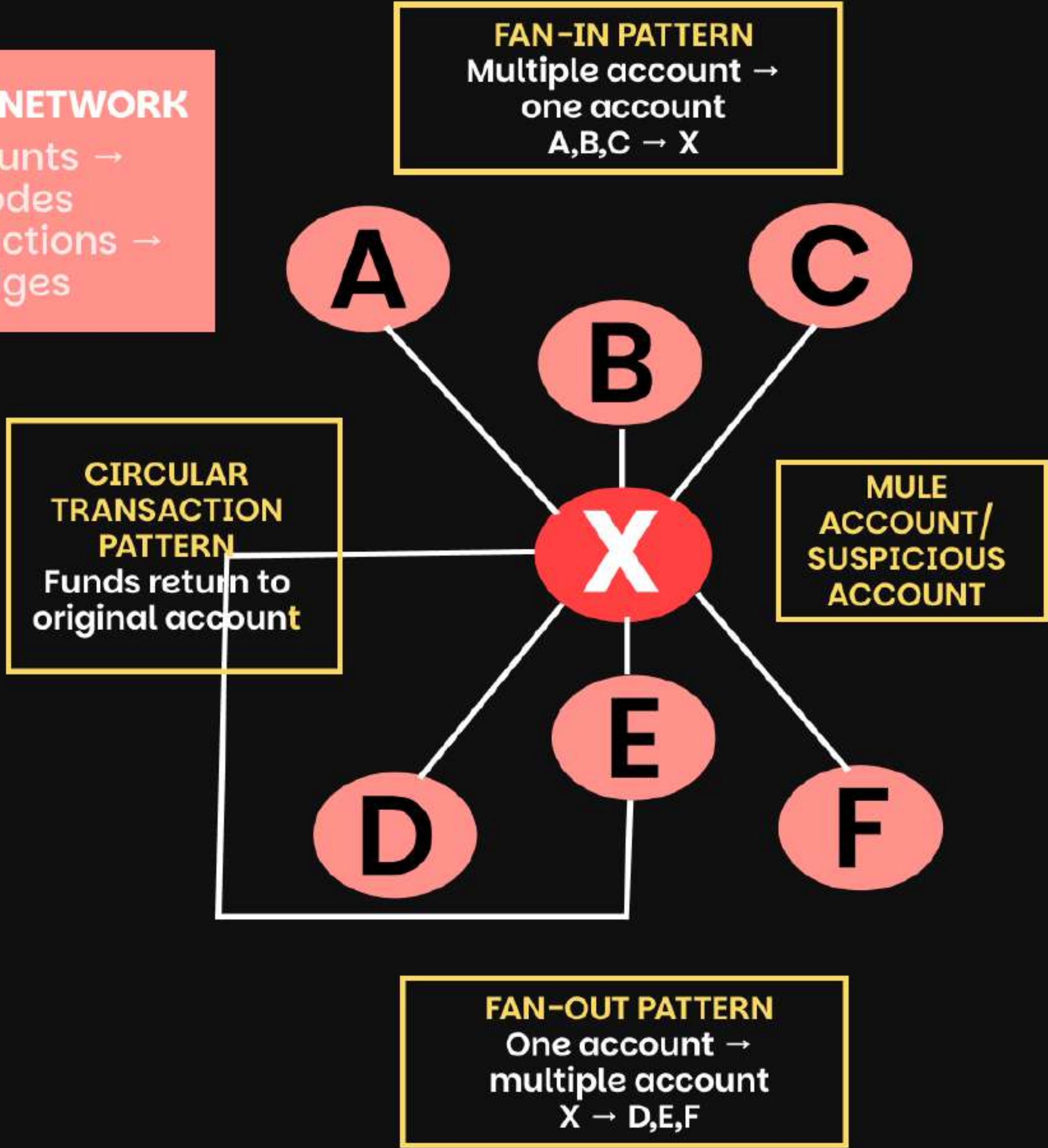
GRAPH ANALYSIS

Financial transaction Network detection

> **Modeling transactions as a network to detect hidden money laundering structures and suspicious account behavior.**

SCALE OF NETWORK

- Accounts → Nodes
- Transactions → Edges



GRAPH METRICES

- In-degree(X) = High
- Out-degree(X) = High
- Centrality(X) = High

Meaning: X is a high-risk node

RISK SCORING ENGINE

Prioritizing High-Risk Accounts

INPUT FEATURES

FEATURE	PURPOSE
Transaction count	Activity level
Total Amount sent	Money movement
Fan-in	Collection detection
Fan-out	Layering Detection
Centrality	Network Importance
Cycle count	Circular Transfer
Night transaction	Time based risk

RISK CLASSIFICATION

0--------30--------60--------80--------100

LOW MEDIUM HIGH CRITICAL

RISK SCORE CALCULATION

Transaction Count +20
Fan-Out Score +15
Fan-In Score +15
Cycle Detection +25
Night Transactions +10

Risk Score = 85

Account X
Risk Score: 85/100
Status: HIGH RISK 🔴

EXPLAINABLE AI

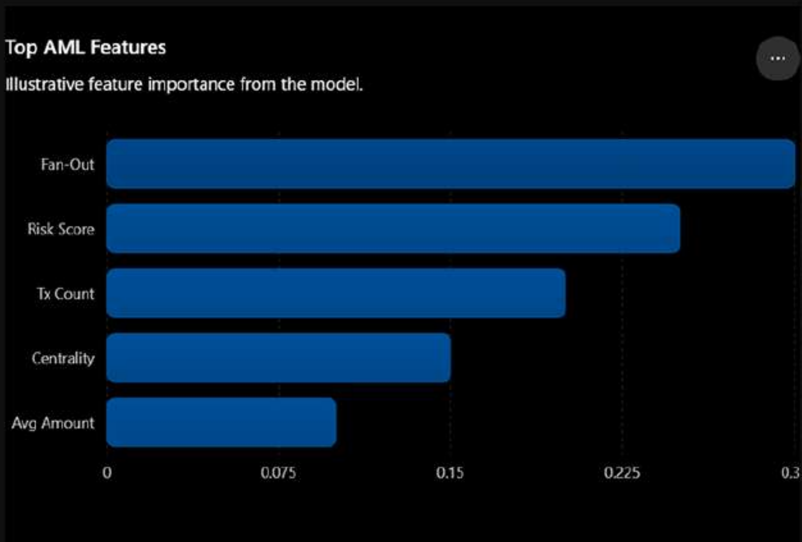
Why was x flagged?

- High Transaction Frequency
- High Fan-Out Pattern
- Circular Transactions Detected
- Night-Time Activity

AML predictions results and risk insights

EXAMPLE

FEATURE	IMPORTANCE
Fan-out score	0.3
risk score	0.25
Total count	0.2
Centrality	0.15
AVG row	0.1



🚨 **AML ALERT**

👤 Account A123

85/100
🔴 CRITICAL

ML Confidence 94%

EVIDENCE
- 🚩 High Fan-Out
- 🚩 Circular Transactions
- 🚩 High Transaction Count

SHAP like explaination

FEAT	IMPORT
Fan-out score	0.3
risk score	0.25
Total count	0.2
Centrality	0.15
AVG row	0.1

AI INVESTIGATION REPORT

This account demonstrates suspicious behavior consistent
with layering activity.

Large fan-out patterns and circular transaction flows significantly increase risk.

EXPLAINABLE AI (XAI)

Understanding model decisions

OBJECTIVE

" Provide transparent and evidence-based explanations for every suspicious account flagged by the AML system "

TRADITIONAL AI

Account A
Risk = 92
Prediction = Laundering

❌ No explanation
❌ Difficult to trust
❌ Not suitable for audits

EXPLAINABLE AI

Account A
Risk Score = 92
ML Confidence = 94%

REASONS:

1. High Fan-Out
2. Circular Transactions
3. High Transaction Frequency
4. High Network Centrality

DEPLOYMENT AND PRODUCTIONIZATION

DEPLOYMENT PLATFORM

Streamlit Cloud / Render
Live Web Application
Accessible via Browser

BUSINESS IMPACT

1.Detect suspicious accounts
2.Prioritize investigations
3.Explain AI decisions
4.Generate automated reports

SYSTEM ARCHITECTURE

FRONTEND LAYER

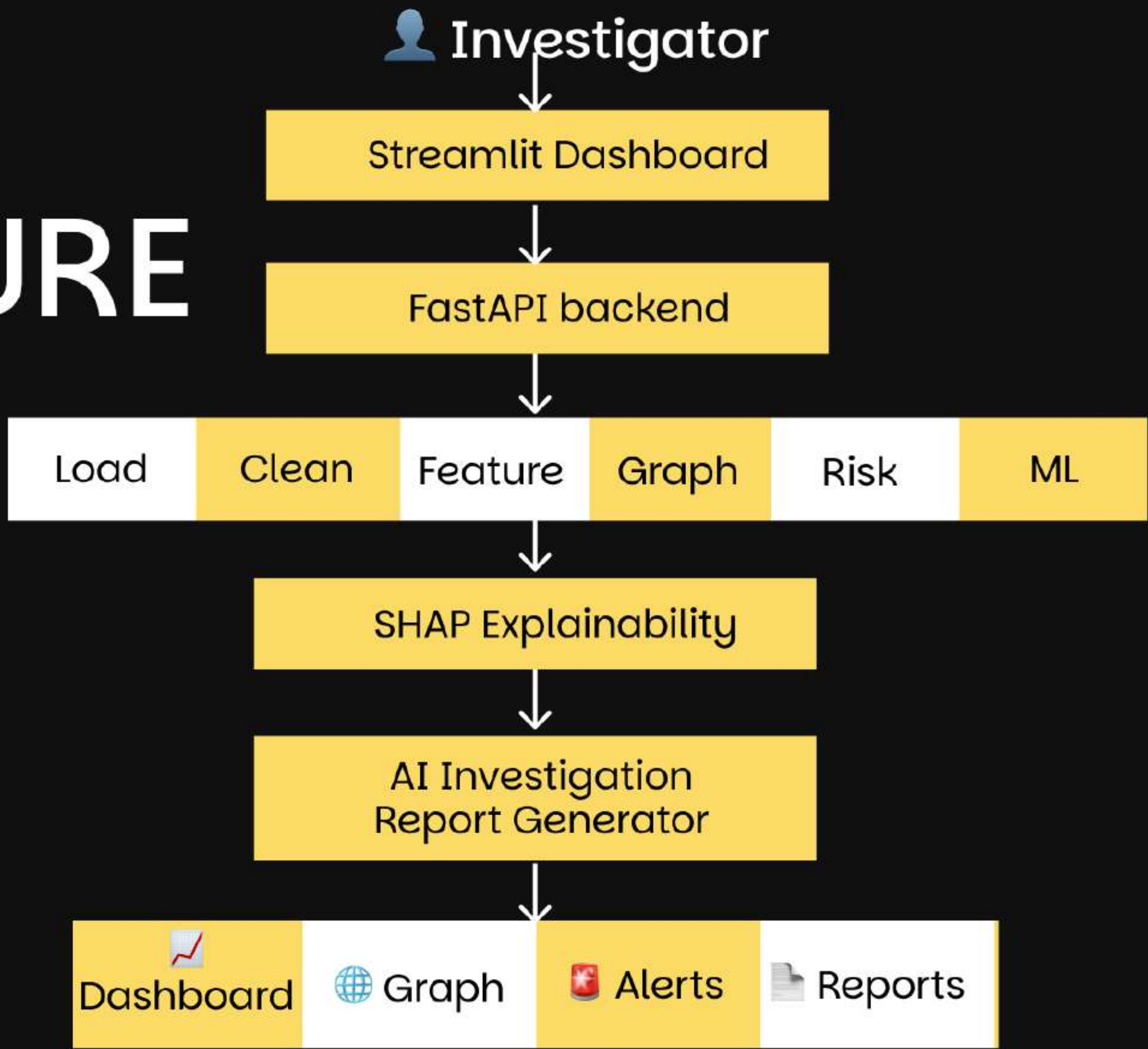
Streamlit Dashboard (User Interface)

OUTPUT LAYER

1. Risk Score
2. Alerts
3. Investigation Reports
4. Network Graphs

BACKEND + AI ENGINE

1. FastAPI
2. Data Processing
3. Feature Engineering
4. Graph Analytics
5. ML Model
6. SHAP Explainabilit
7. Risk Engine



TECH STACK USED AML DETECTION

Frontend (User Interface)

Streamlit

Backend (API Layer)

Flask API

Machine Learning

Skicit learn
XG boot

Data Processing & Analysis

Pandas
Numpy

Graph Analytics (Core AML Feature)

NetworkX
Pyvix

Explainable AI (XAI)

SHAP

Data Visualization

Matplotlib
Seaborn

Data Storage

CSV/Excel
Postgree SQL (Advanced)

Deployment

Streamlit cloud
Heroku
Render